# VOLODYMYR POLISHCHUK

Head of Software & Autonomy | Systems Architect | GPS-Denied UAV Navigation

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## SUMMARY

Head of software and autonomy and systems architect with 19 years of designing production systems at scale, from Wix's developer platform to Playtech's real-time betting engine and UkrCard's payment network. Since 2026 I have built the software and autonomy function of Applied Technology Collective, a Ukrainian UAV developer, from zero: GPS-denied navigation now in production, with its own flight-controller firmware and simulator, and the team behind it. I work from first principles, breaking a problem down to its physical, mathematical and computational realities instead of adopting frameworks by analogy. I build AI-native, owning the architecture, the mathematics and the acceptance criteria and directing AI coding agents as the implementation team.

## EXPERIENCE

### Applied Technology Collective | Head of Software & Autonomy

**Kyiv, Ukraine | 2026 – Present**

Ukrainian developer of strike, loitering-munition and cargo UAVs. Joined as the only software engineer, built the software and autonomy function from zero and hired its first two engineers. NADIR, SEXTANT and ORRERY are my own IP, used by the company; code walkthrough on request.

**NADIR — GPS-Denied Position Hold (In Production)**

- Freed the company's drones from depending on GPS to hold position with visual-inertial navigation: one downward camera and the autopilot's IMU give the velocity and position ArduPilot flies on in place of satellites.
- Kept the hardware to $15 per drone, shipped as a sealed product that operators calibrate in the field and that refuses to arm until the setup is safe.

**SEXTANT — Long-Range Terrain Navigation (Rolling Out)**

- Extended GPS-denied flight to long range by fixing absolute position and heading from the terrain's geometry, matched against a terrain model, so that navigation survives the snow, foliage and changing light that defeat image matching.

**ORRERY — Standalone Flight Simulator**

- Cut the cost and risk of flight testing with a simulator in which NADIR and SEXTANT pass their acceptance tests before reaching an aircraft; any ArduPilot SITL build, camera and vehicle plug in through configuration alone.

### Wix.com | Principal Software Engineer / Architect

**Kyiv, Ukraine | 2017 – 2026**

**Platformization & Engineering Culture**

- Laid the foundation for opening Wix's internal APIs to third-party developers and the App Market by leading the company-wide adoption of one API contract standard, enforced in the pipeline every service passes.

- Moved the backend to asynchronous code and cut its resource costs through courses and workshops I designed and taught, mandatory for Wix's 1,000+ server engineers.
- Shaped Wix's engineering hiring through 300+ interviews and a vote on the hiring committees.

**Platforms & Infrastructure**

- Took infrastructure off the hands of about 3,000 developers across 2,000+ microservices by architecting Wix's internal serverless platform, built with a core team of five and scaling without downtime.
- Brought AI assistance to the Wix App Market by architecting its retrieval-augmented chat agent.
- Made Wix independent of third-party registrars with its own ICANN-accredited registrar and DNS, automated SSL for 8M+ domains, and led domain management's move from a legacy monolith to microservices.

### Playtech | Software Architect / Tech Lead

**Kyiv, Ukraine | 2012 – 2017**

- As tech lead of a team of five (a team leader handled management), set the design, technical direction and goals, and led the betting platform's migration from a monolith to a reactive architecture.
- Held 60M requests/min peaks during the FIFA World Cup with real-time streaming, on a highly available 3-node minimum that cut hardware costs.
- Kept the platform within regulation that ruled out the cloud by designing a distributed key-value store on bare metal that serves reads locally.

### UkrCard | Systems Architect

**Kyiv, Ukraine | 2007 – 2012**

- Secured the payment network by leading the move of every ATM, POS and self-service terminal from wired X.25 to encrypted wireless TCP/IP.
- Cut costly false positives in fraud detection with a neural-network assistant that learned from operator feedback (Master's project).

## SKILLS

**Autonomy:** visual-inertial navigation, sensor fusion, terrain-referenced navigation, ArduPilot, MAVLink.
**Engineering:** Rust, C, Scala, Java, Python; embedded Linux, distributed systems, Kubernetes, gRPC.
**AI:** RAG, LLM applications, neural networks from first principles.

## EDUCATION

**M.Sc. in Cybernetics** | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics. Thesis: Ensuring the Dependability of Transaction Classification in a Payment System.

**B.Sc. in Applied Mathematics** | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics

## MILITARY & OTHER QUALIFICATIONS

**Armed Forces of Ukraine (reserve):** Junior Lieutenant, military specialty 708200: operation, tactical deployment and maintenance of air-defense radar systems.
**IYT Skipper** (International Yacht Training): practical navigation.
**Interests:** world models for robot navigation (V-JEPA, Genesis); basal cognition (Michael Levin); foundations of physics.